

A Packing of 5 Four-Spaces in $\mathbb{R}^9$

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Abstract

We give 5 4-dimensional subspaces of $\mathbb{R}^9$ whose smallest squared chordal distance is at least 2.777625. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.666497, obtained by deleting 1 of the 6 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 5 subspaces of dimension 4 in $\mathbb{R}^9$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.777625 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 9$, $n = 4$. The cell $N = 5$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.666497, obtained by deleting 1 of the 6 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(9 - 4)/9 \cdot 5/4 = 2.777778$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m9n4N5.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
115734406242723138661207085015757397137190096698543120258779606798014218
580115830347713204991401876267835843282439861409378754859098856541735669
446518657703839071037628473593831764015742644867
```

denominator:

```
416666666666229873356556685136159078400234858702373922760971070100945851
392334678086623730098603250829516827959522079811525386689092207340874990
89140768061565660655361725497223969161297129934
```

B Coordinates

5 blocks of 9 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m9n4N5.txt`.

```
-175917973791 -322828938726 -271072839421 19544819662302095447 519991093865 -624376492834 -147297016552816076394 -481910645999 155855583657 496410293245
-469663971422 268920734147 -513135455992 348989231377365109821 -289024341333 142849821974 12167245982221703971941 -99240143638 -527662777636 531466581394
479808716633 21487624922 -579848025353 -16505457896681974561932 322469885540 397324934517 -831934433505161656088 166091350524 174264661512 242292091921
332707495925 -229416001143 -316351220839 6742325925259675650113 -398500380521 160034782110 -189730878394275096104 216131951913 310800592189 -93836679960
-81514614064 -349894909184 -221182788003 -122625295738289006799 641845711140 -1118414980 219951540642-92525150086 -37129582630 560256587059 107042922297
-321619964680 -353974472927 259205442702 5027488085345277484229 233375428078 553046965748 216999392647-306378949557 315483113216 214478115687 492326938287
232095943567 -38316153112 265807692121 -26301821365711492698307 -268382758455 -590341091316 24336094033830567555 192174965431 55443256254 63742027916
-160523124076 -684021027017 -83301732999 -425100219157083401765 -369972870238 148028899263 23378319390-345897619112 -85819656429 -339888459203 -43432014616
466730868567 -230851526991 182229242861 9545320174 139279953858 99854908126 -41744386210 -698318699394-91894631977 -110083481404 384342642084 260071638273
-558449527575 -280185679371 -250121378540 3022151894188025800619 166303343675 -435604525227 -900560374209092262105 -255683049517 -549819663226 -222116541014
-213139518231 -389860235970 -102072213328 48281052343904372867 342138187345 -304361493042 -15148732303623827453491 158742489433 137383093712 631337028718
-124348985542 -318902757583 -365499334777 -470264072393297188123 105697046328 118244228620 -53856863914220728108529 700047339063 -24985520120 5004636869
134379573396 -66508776635 105687907486 -119552318450449960535974 -455511902843 409260523642 -120556643454305024948 -214529346439 -431110821187 660054830987
204831808865 335016715554 -280466783136 60840238329616769766490 520663377006 184979223464 27651642356387749704002 -486704497581 246218428805 113426363320
22491791470 -305342039293 -368076167768 -162922393382122871312 -322683311297 93359814734 242425398721-389892444196 -282916968762 406697069772 -172997656434
```